

PPO Summary: From Policy Gradient Theorem to State-of-the-Art GAE Implementation

This document provides a concise summary of the 8-step mathematical journey from the foundational policy gradient theorem to the sophisticated Generalized Advantage Estimation (GAE) used in modern PPO implementations.

Step 1: Policy Gradient Foundation

Core Objective: Maximize expected return $J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[R(\tau)]$

Key Assumptions:

- MDP Framework: States $\mathcal{S}$, actions $\mathcal{A}$, transitions $P(s'|s, a)$, rewards $R(s, a)$, discount γ
- Differentiability: Policy $\pi_\theta(a|s)$ differentiable w.r.t. θ
- Stationarity: Time-invariant dynamics and policy (momentarily during gradient computation)

Policy Gradient Theorem:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) R_t \right]$$

Key Equations:

- Return: $R_t = \sum_{k=t}^{T-1} \gamma^{k-t} r_k$ (causality principle)
- Likelihood ratio trick: $\nabla_\theta \mathbb{E}_{x \sim p_\theta}[f(x)] = \mathbb{E}_{x \sim p_\theta}[f(x) \nabla_\theta \log p_\theta(x)]$

Rationale: Uses likelihood ratio trick and causality principle to derive unbiased gradient estimator. Log probability provides numerical stability, natural gradient interpretation, and mathematical elegance.

Step 2: The High Variance Problem and Baseline Subtraction

Problem: Basic policy gradient suffers from extremely high variance, making learning unstable.

Key Concepts:

- **Variance:** Gradient estimates fluctuate around expected value across trajectory samples
- **Bias:** Systematic error between estimator expectation and true gradient

Solution - Baseline Subtraction Theorem:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) (R_t - b(s_t)) \right]$$

Key Insight: $\mathbb{E}_{a \sim \pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(a | s) b(s)] = 0$

Optimal Baseline: $b^*(s) \approx V^{\pi}(s) = \mathbb{E}_{a \sim \pi_{\theta}} [Q^{\pi}(s, a)]$

Rationale: Any state-dependent baseline can be subtracted without introducing bias, with state value function being the variance-minimizing choice.

Step 3: Emergence of Advantage Functions

Mathematical Derivation: Using $V^{\pi}(s)$ as baseline leads to advantage functions:

$$A^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s)$$

Advantage-Based Policy Gradient:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) A^{\pi}(s_t, a_t) \right]$$

Critical Property: $\mathbb{E}_{a \sim \pi_{\theta}} [A^{\pi}(s, a)] = 0$ (zero-mean)

Mathematical Proof:

$$\mathbb{E}_{a \sim \pi_{\theta}} [A^{\pi}(s, a)] = \mathbb{E}_{a \sim \pi_{\theta}} [Q^{\pi}(s, a)] - V^{\pi}(s) = V^{\pi}(s) - V^{\pi}(s) = 0$$

Rationale: Advantages provide relative action quality information without introducing bias, ensuring unbiased policy gradient updates while reducing variance.

Step 4: Temporal Difference (TD) Estimation

Problem: Direct computation of $Q^{\pi}(s, a)$ and $V^{\pi}(s)$ is intractable.

TD Solution: Use Bellman equation to estimate advantages:

$$\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$$

Key Assumptions:

- Bellman Validity: $V^\pi(s_t) = \mathbb{E}_\pi[r_t + \gamma V^\pi(s_{t+1}) | s_t]$
- Markov Property: Future depends only on current state
- Value Function Accuracy: $V(s) \approx V^\pi(s)$

Bias-Variance Tradeoff:

- **Monte Carlo Direction:** Low bias, high variance (using actual rewards)
- **TD Direction:** High bias, low variance (using bootstrapped estimates)

Advantage Approximation: When $V(s) = V^\pi(s)$, then $\delta_t^V = A^\pi(s_t, a_t)$

Rationale: TD errors provide computationally tractable advantage estimates, introducing the fundamental bias-variance tradeoff in advantage estimation.

Step 5: The Bias-Variance Tradeoff in Advantage Estimation

Core Dilemma:

- **TD Error ($\lambda = 0$):** $\hat{A}_t = \delta_t \rightarrow$ High bias, Low variance
- **Monte Carlo ($\lambda = 1$):** $\hat{A}_t = \sum_{k=0}^{\infty} \gamma^k \delta_{t+k} \rightarrow$ Low bias, High variance

High Variance Sources:

1. Stochastic rewards and environment transitions
2. Function approximation errors
3. Finite sampling

Bias Sources:

1. Value function approximation: $V_\phi(s) \neq V^\pi(s)$
2. Bootstrapping with estimated values
3. Finite horizon truncation

Rationale: Understanding the sources of bias and variance enables principled design of advantage estimators that balance these competing objectives.

Step 6: Generalized Advantage Estimation (GAE) - The Optimal Solution

Mathematical Foundation:

$$\hat{A}_t^{GAE(\gamma,\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}^V$$

Recursive Implementation:

$$\hat{A}_t^{GAE} = \delta_t + \gamma\lambda\hat{A}_{t+1}^{GAE}$$

Bias-Variance Control Parameter λ :

- $\lambda = 0$: Pure TD $\rightarrow$ High bias, low variance
- $\lambda = 1$: Pure MC $\rightarrow$ Low bias, high variance
- $\lambda \in (0, 1)$: Optimal tradeoff $\rightarrow$ Exponentially weighted combination

Key Assumptions:

1. Exponential decay: Future information becomes less reliable exponentially
2. Geometric convergence: $|\gamma\lambda| < 1$ ensures series convergence
3. Finite horizon: Practical truncation at episode boundaries

Rationale: GAE provides principled interpolation between bias and variance, allowing practitioners to tune the λ parameter for optimal performance in specific domains.

Step 7: State-of-the-Art Implementation in PPO

Practical GAE Algorithm:

```
import jax.numpy as jnp

def gae_advantages(rewards, terminal_masks, values, discount, gae_param):
    advantages = []
    gae = 0.0
    for t in reversed(range(len(rewards))):
        delta = rewards[t] + discount * values[t + 1] * terminal_masks[t] - values[t]
        gae = delta + discount * gae_param * terminal_masks[t] * gae
        advantages.append(gae)
    return jnp.array(advantages[::-1])
```

Why Reversed Loop is Essential:

- **Mathematical Dependency:** $\hat{A}_t^{GAE} = \delta_t + \gamma\lambda\hat{A}_{t+1}^{GAE}$ creates backward dependency chain
- **Boundary Condition:** $\hat{A}_T^{GAE} = \delta_T$ (terminal condition)

- **Computational Efficiency:** $O(T)$ linear time complexity
- **Terminal State Handling:** Prevents advantage bleeding across episodes

Implementation Assumptions:

- Neural network approximation: $V_\phi(s) \approx V^\pi(s)$
- Backward processing for efficient computation
- Terminal state handling: $V(s_{\text{terminal}}) = 0$
- Batch processing across parallel environments

Hyperparameter Sensitivity:

- $\gamma \in [0.95, 0.99]$: Discount factor
- $\lambda \in [0.9, 0.99]$: GAE parameter

Rationale: Efficient implementation that maintains mathematical correctness while enabling practical deployment in modern RL systems.

Step 8: Integration with PPO Loss Function

Final Policy Gradient with GAE:

$$\nabla_\theta J(\theta) \approx \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t | s_t) \hat{A}_t^{GAE} \right]$$

PPO Clipped Objective:

$$L^{CLIP}(\theta) = \mathbb{E}_t[\min(r_t(\theta) \hat{A}_t^{GAE}, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t^{GAE})]$$

where $r_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)}$ is the importance sampling ratio.

Key Components:

1. **Importance Sampling Ratio:** Enables off-policy updates and sample reuse
 - Corrects for policy mismatch between data collection and updates
 - Mathematical foundation:

$$\mathbb{E}_{\tau \sim \pi_\theta}[f(\tau)] = \mathbb{E}_{\tau \sim \pi_{\theta_{old}}} \left[\frac{\pi_\theta(\tau)}{\pi_{\theta_{old}}(\tau)} f(\tau) \right]$$

PPO's Hybrid On-Policy/Off-Policy Approach:

The use of importance sampling in PPO represents a **perfect example** of the on-policy/off-policy discussion in reinforcement learning, creating a sophisticated hybrid approach:

Traditional Distinction:

- **On-Policy Methods:** Learn about the policy being used to make decisions (e.g., REINFORCE, SARSA)
 -  Theoretical guarantees
 -  Stable training
 -  Sample inefficient
- **Off-Policy Methods:** Can learn from data generated by different policies (e.g., Q-learning, DQN)
 -  Sample efficient
 -  Can learn from any data
 -  Can be unstable
 -  Distribution mismatch issues

PPO's Innovation: PPO bridges this gap through its two-phase process:

- **Experience Collection:** Use policy $\pi_{\theta_{old}}$ (on-policy data collection)
- **Multiple Updates:** Reuse same data for several gradient steps (off-policy learning with importance sampling correction)

The Fundamental Tradeoff PPO Solves:

Traditional On-Policy: $\pi_{\theta_{old}} \rightarrow \text{collect data} \rightarrow \text{one update} \rightarrow \pi_{\theta_{new}} \rightarrow \text{discard data}$

PPO Approach: $\pi_{\theta_{old}} \rightarrow \text{collect data} \rightarrow \text{multiple updates} \rightarrow \pi_{\theta_{new}}$ (reuse same data!)

Why This Matters:

- **Sample Efficiency:** Can reuse data multiple times (like off-policy methods)
- **Stability:** Clipping prevents destructive updates (like on-policy methods)
- **Practical Performance:** Works well across diverse domains
- **Theoretical Soundness:** Maintains convergence guarantees

This hybrid approach demonstrates that the on-policy/off-policy distinction isn't binary but rather a spectrum, with PPO finding an optimal balance between sample efficiency and training stability.

2. Clipping Mechanism: Prevents destructively large policy updates

$$\text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) = \begin{cases} 1 - \epsilon & \text{if } r_t(\theta) < 1 - \epsilon \\ r_t(\theta) & \text{if } 1 - \epsilon \leq r_t(\theta) \leq 1 + \epsilon \\ 1 + \epsilon & \text{if } r_t(\theta) > 1 + \epsilon \end{cases}$$

3. Complete PPO Loss:

$$L_{PPO}(\theta) = L^{CLIP}(\theta) - c_1 L^{VF}(\theta) + c_2 S[\pi_\theta]$$

Theoretical Properties:

- Conservative updates through clipping
- Monotonic improvement guarantees

- Sample efficiency via experience reuse
- Training stability through bounded policy changes

Rationale: Integration of GAE with PPO's clipped objective creates a theoretically sound and practically effective algorithm that balances sample efficiency, training stability, and performance.

Summary of Key Benefits

1. **Variance Reduction:** GAE smooths noisy advantage estimates
2. **Bias Control:** λ parameter enables fine-tuning of bias-variance tradeoff
3. **Sample Efficiency:** Better gradient estimates accelerate learning
4. **Stability:** Reduced gradient noise and clipping improve training stability
5. **Credit Assignment:** Multi-horizon consideration improves temporal credit assignment

Theoretical Guarantees

- **Unbiasedness:** When $V(s) = V^\pi(s)$, GAE provides unbiased advantage estimates
- **Consistency:** As $\lambda \rightarrow 1$, GAE approaches Monte Carlo estimates
- **Convergence:** Maintains policy gradient theorem guarantees while improving practical performance

This progression represents a principled evolution driven by the fundamental bias-variance tradeoff, culminating in state-of-the-art advantage estimation used in modern PPO implementations.

Practical Tensor Structures for PPO Optimization

This section ties the 8-step conceptual journey to the concrete tensors that flow into the PPO loss in this NNX-based implementation.

- Collection across parallel environments and time:
 - We run N parallel simulators for T steps to collect experience.
 - Raw stacks (before flattening):
 - states: shape $(T, N, 84, 84, 4)$
 - actions: shape (T, N) (discrete action indices)
 - rewards: shape (T, N)
 - values: shape $(T+1, N)$ (bootstrap one extra value for GAE)
 - old_log_probs: shape (T, N) ($\log \pi_{\{\theta_{old}\}}(a_{t|s_t})$)
 - dones: shape (T, N) (1.0 at terminal, else 0.0)
- Per-environment GAE and returns (Step 6–7):

- For each environment $i \in \{1, \dots, N\}$, form terminal masks $m_t = 1 - \text{done}_t$.
- Call GAE with vectors of length T :
 - $\text{rewards}[:, i] \rightarrow (T,)$
 - $\text{terminal_masks} \rightarrow (T,)$
 - $\text{values}[:, i] \rightarrow (T+1,)$
- Compute for each env i :
 - $\text{advantages}[:, i]$: shape $(T,)$
 - $\text{returns}[:, i] = \text{advantages}[:, i] + \text{values}[:, -1, i]$: shape $(T,)$
- Flatten for optimization (Step 8):
 - Define $B = T \times N$ (the total number of time-steps across all envs).
 - The training trajectories passed to the optimizer are flattened to:
 - states : $(B, 84, 84, 4)$
 - actions : $(B,)$ int32
 - old_log_probs : $(B,)$ float32
 - returns : $(B,)$ float32
 - advantages : $(B,)$ float32 (normalized inside the loss)
- Minibatching for the loss:
 - During an epoch, the flattened arrays are reshaped to $(\text{iterations}, \text{batch_size}, \dots)$ and iterated.
 - The loss function consumes one minibatch tuple:
 - states : $(\text{batch_size}, 84, 84, 4)$
 - actions : $(\text{batch_size},)$
 - old_log_probs : $(\text{batch_size},)$
 - returns : $(\text{batch_size},)$
 - advantages : $(\text{batch_size},)$
 - Model outputs for discrete actions:
 - log_probs : $(\text{batch_size}, \text{num_actions})$
 - values : $(\text{batch_size}, 1) \rightarrow$ squeezed to $(\text{batch_size},)$
- What the PPO loss computes per minibatch (high level):
 - Gather $\text{log_probs_act_taken} = \text{log_probs}[j, \text{actions}[j]]$ per sample.
 - Ratio $r_t = \exp(\text{log_probs_act_taken} - \text{old_log_probs})$.
 - Normalize advantages: $(A - \text{mean}(A)) / (\text{std}(A) + 1e-8)$.
 - Policy loss = $-\text{mean}(\min(r_t \cdot A, \text{clip}(r_t, 1-\epsilon, 1+\epsilon) \cdot A))$.
 - Value loss = $\text{mean}((\text{returns} - \text{values})^2)$.
 - Entropy bonus = mean categorical entropy of the policy.

Concrete examples

1. Typical Atari-like setup (discrete actions)

- Suppose $T = 128$ rollout steps and $N = 8$ environments, so $B = 1024$.

- `num_actions = 4` (example).
- After flattening:
 - `states.shape = (1024, 84, 84, 4)`, `dtype=float32`
 - `actions.shape = (1024,)`, `dtype=int32`, e.g. `[1, 0, 3, 2, 1, ...]`
 - `old_log_probs.shape = (1024,)`, e.g. `[-0.69, -1.20, -0.92, ...]`
 - `returns.shape = (1024,)`, e.g. `[0.43, 0.51, -0.12, ...]`
 - `advantages.shape = (1024,)`, normalized to `mean≈0` and `std≈1`
- A single minibatch with `batch_size = 256` would have the same per-field shapes with 256 instead of 1024.

2. Tiny end-to-end toy snapshot ($T = 2$, $N = 3 \Rightarrow B = 6$)

- Flattened tensors passed into one loss evaluation (`batch_size = 6`):
 - `states`: `(6, 84, 84, 4)`
 - `actions`: `[2, 0, 1, 3, 1, 0]`
 - `old_log_probs`: `[-1.20, -0.69, -1.10, -1.39, -0.75, -1.25]`
 - `returns`: `[0.97, 0.12, -0.05, 0.22, 0.08, -0.11]`
 - `advantages` (before normalization): `[0.50, -0.10, 0.20, -0.30, 1.00, -1.30]`
 - `advantages` (after normalization inside loss): `mean≈0`, `std≈1`
- The model produces:
 - `log_probs`: `(6, num_actions)`
 - `values`: `(6, 1) → (6,)`
- The loss then computes `r_t`, applies clipping with ϵ , value MSE to returns, and adds an entropy bonus.

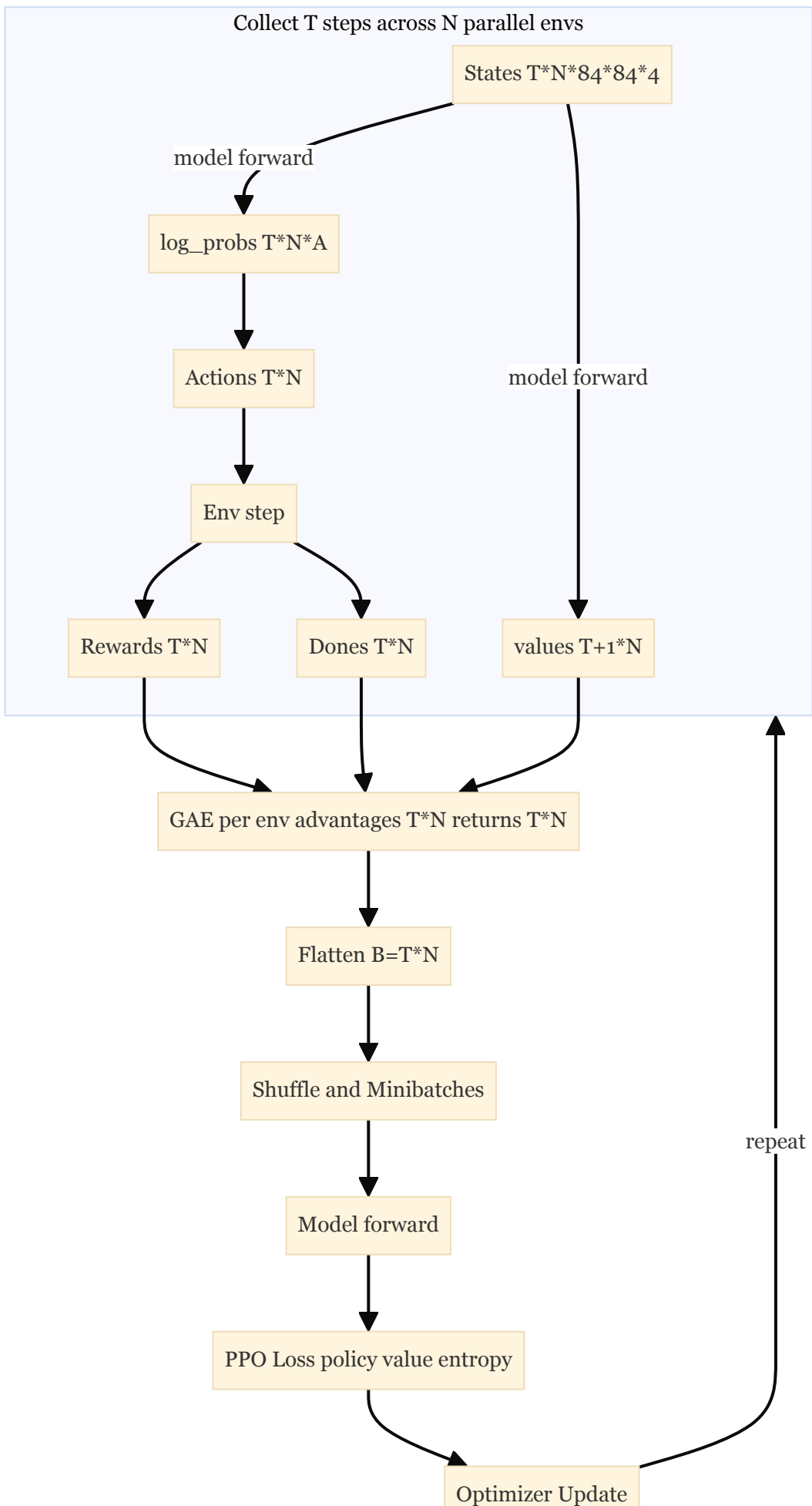
Notes and edge cases

- Terminal handling: masks prevent advantages from leaking across episode boundaries.
- Dtypes: actions are integer indices; all other scalars are `float32`.
- Continuous action variants would use actions shaped `(B, action_dim)` and log-prob scalars per sample, but this example uses discrete actions (categorical policy).

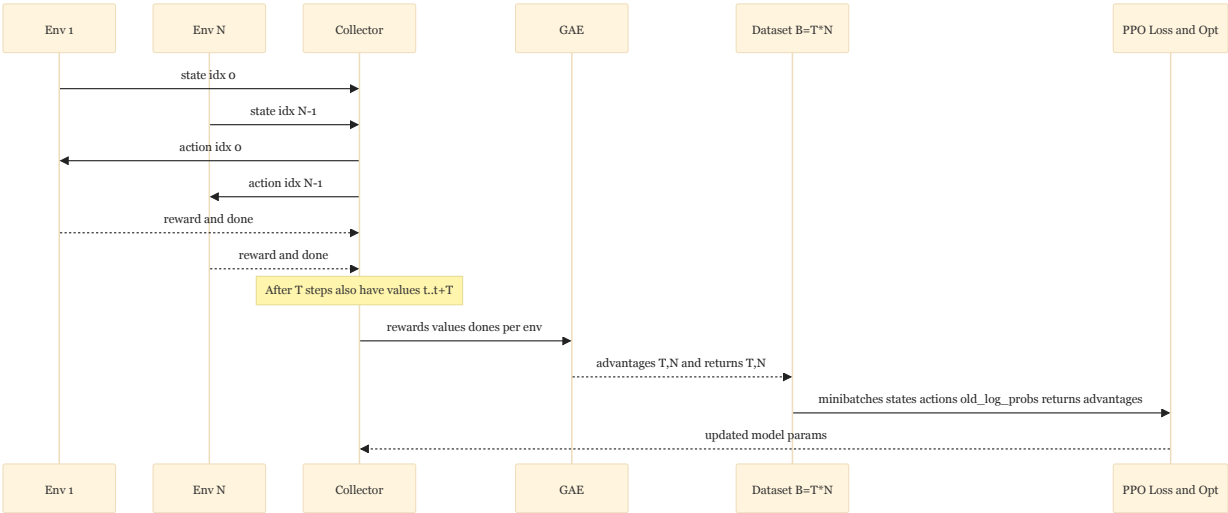
Visualizing the PPO Data Process

A concise way to understand how tensors move through PPO is to visualize the pipeline from rollout collection to the optimized update.

Flowchart overview



End-to-end interaction (sequence)



What to plot while debugging or explaining the flow

- Advantages histogram: expect mean≈0 after normalization, track std.
- Policy ratio `r_t` histogram and clip fraction: how often clipping is active.
- Value loss and returns vs. predicted values scatter: calibration of critic.
- Entropy and action distribution: policy exploration over time.
- Episode length/reward curves: aligns with data segmentation and masks.

Tip: These can be logged with TensorBoard; this repo already writes scalar `game_score`. Adding histograms for advantages and ratios in the training loop can make these plots immediately available.